



Robotic Systems

Chapter V Inverse Kinematics

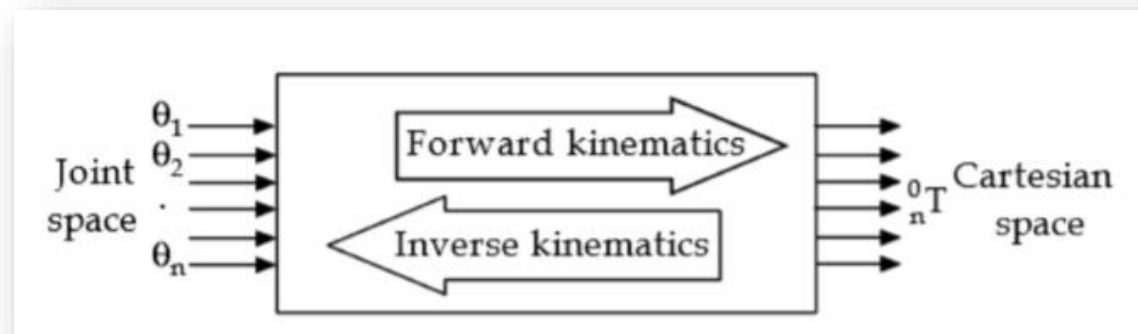
Prof. Dr. Ibrahim Hamarash
University of Kurdistan-Hewlêr

Inverse Kinematics

The Inverse Kinematics can be described as follows:

Given the desired position and orientation of the end effector frame relative to a reference frame,

how do we compute the set of joint angles which will achieve this desired result?



Inverse Kinematics, mathematically

The general problem of inverse kinematics can be stated as follows. Given a 4 x 4 homogeneous transformation

$$H = \begin{bmatrix} R & o \\ 0 & 1 \end{bmatrix} \in SE(3)$$

Find a solution, or possibly multiple solutions, of the equation

$$T_n^0(q_1, \dots, q_n) = H$$

Solving IK problem:

Solving the problem of finding the required joint angles to place the end effector frame $\{W\}$, relative to the reference frame $\{S\}$, is split into two parts. First, frame transformations are performed to find the end effector frame $\{W\}$, relative to the reference frame $\{S\}$,

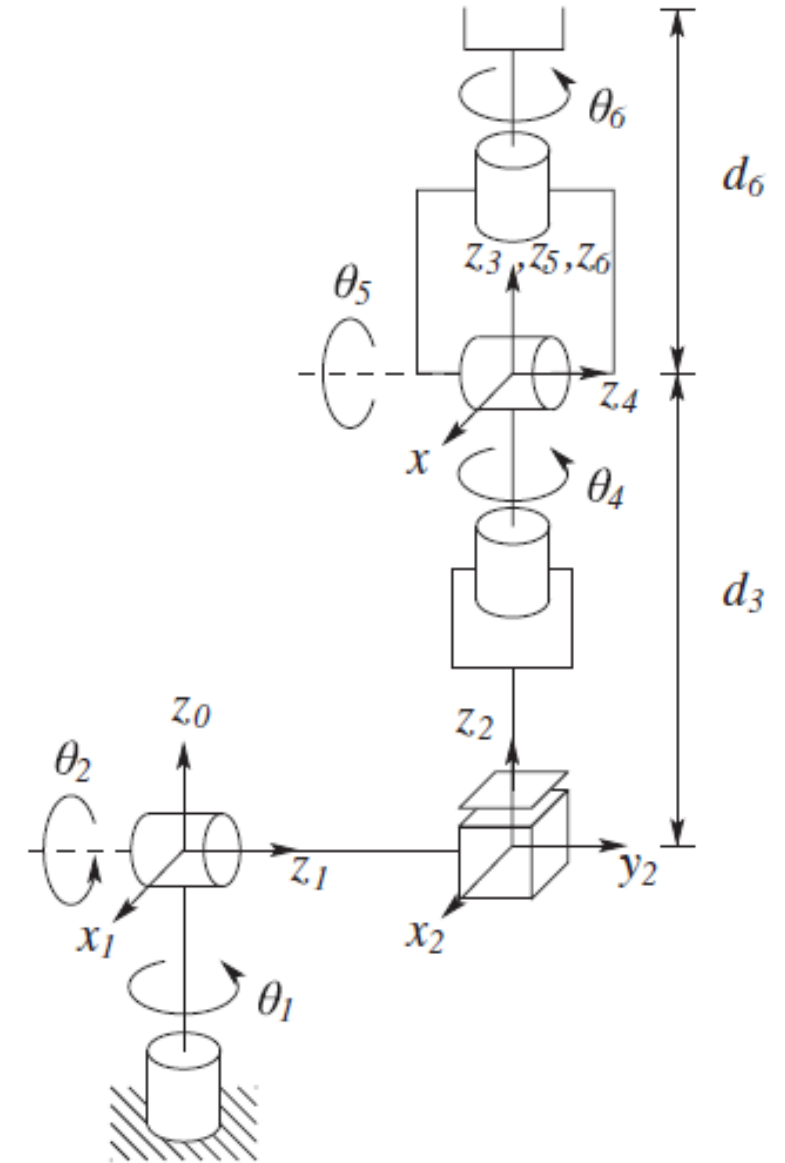
and then the inverse kinematics are used to solve for the joint angles.

The problem of solving the kinematic equations of a manipulator is a nonlinear one.

Given the numerical value of 0_nT , we attempt to find values of $\theta_1, \theta_2, \dots, \theta_n$.

Stanford Arm (Puma 560): DH Link Parameters

Link	d_i	a_i	α_i	θ_i
1	0	0	-90	θ_1
2	d_2	0	$+90$	θ_2
3	d_3	0	0	0
4	0	0	-90	θ_4
5	0	0	$+90$	θ_5
6	d_6	0	0	θ_6



Stanford Arm (PUMA 560) Transformation Matrices

$$A_1 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_5 = \begin{bmatrix} c_5 & 0 & s_5 & 0 \\ s_5 & 0 & -c_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_6 = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Stanford Arm (PUMA 560) Transformation Equation

$$T_6^0 = A_1 \cdots A_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & d_x \\ r_{21} & r_{22} & r_{23} & d_y \\ r_{31} & r_{32} & r_{33} & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} r_{11} &= c_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] - s_1(s_4c_5c_6 + c_4s_6) \\ r_{21} &= s_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] + c_1(s_4c_5c_6 + c_4s_6) \\ r_{31} &= -s_2(c_4c_5c_6 - s_4s_6) - c_2s_5c_6 \\ r_{12} &= c_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] - s_1(-s_4c_5s_6 + c_4c_6) \\ r_{22} &= -s_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] + c_1(-s_4c_5s_6 + c_4c_6) \\ r_{32} &= s_2(c_4c_5s_6 + s_4c_6) + c_2s_5s_6 \\ r_{13} &= c_1(c_2c_4s_5 + s_2c_5) - s_1s_4s_5 \\ r_{23} &= s_1(c_2c_4s_5 + s_2c_5) + c_1s_4s_5 \\ r_{33} &= -s_2c_4s_5 + c_2c_5 \\ d_x &= c_1s_2d_3 - s_1d_2 + d_6(c_1c_2c_4s_5 + c_1c_5s_2 - s_1s_4s_5) \\ d_y &= s_1s_2d_3 + c_1d_2 + d_6(c_1s_4s_5 + c_2c_4s_1s_5 + c_5s_1s_2) \\ d_z &= c_2d_3 + d_6(c_2c_5 - c_4s_2s_5) \end{aligned}$$

Inverse Kinematics: PUMA 560

In the case of the PUMA 560 manipulator, the precise statement of Inverse Kinematic problem is as follows:

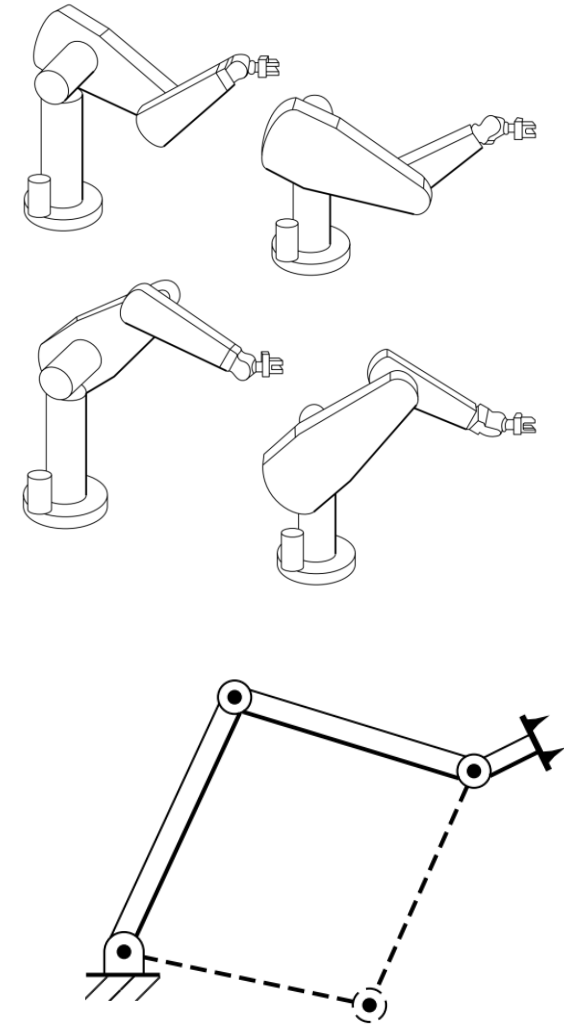
Given 0_6T as sixteen numeric values (four of which are trivial), so we can express the remaining twelve nonlinear equations as:

$$T_{ij}(q_1, \dots, q_n) = h_{ij}, \quad \text{for } i = 1, 2, 3, \quad j = 1, 2, 3, 4$$

Solvability

A manipulator will be considered solvable if the joint variables can be determined by an algorithm that allows one to determine all the sets of joint variables associated with a given position and orientation.

For the case of an arm with six degrees of freedom (like the one corresponding to the Puma 560) we have 12 equations and six unknowns. However, among the 9 equations arising from the rotation-matrix portion of T_{06} , only 3 are independent. These, added to the 3 equations from the position-vector portion of T_{06} , give 6 equations with six unknowns. These equations are nonlinear, transcendental equations, which can be quite difficult to solve.



Existence of solutions

The question of whether any solution exists at all raises the question of the manipulator's workspace. Roughly speaking, workspace is that volume of space that the end-effector of the manipulator can reach. For a solution to exist, the specified goal point must lie within the workspace. The reachable workspace is that volume of space that the robot can reach in at least one orientation.

Example

Suppose that the desired position and orientation of the final frame are given by

$$H = \begin{bmatrix} 0 & 1 & 0 & -0.154 \\ 0 & 0 & 1 & 0.763 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

If the values of the nonzero DH parameters are $d_2=0.154$ and $d_6=0.263$, one solution to this set of equations is given by:

$$c_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] - s_1(s_4c_5c_6 + c_4s_6) = 0$$

$$s_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] + c_1(s_4c_5c_6 + c_4s_6) = 0$$

$$-s_2(c_4c_5c_6 - s_4s_6) - c_2s_5c_6 = 1$$

$$c_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] - s_1(-s_4c_5s_6 + c_4c_6) = 1$$

$$s_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] + c_1(-s_4c_5s_6 + c_4c_6) = 0$$

$$s_2(c_4c_5s_6 + s_4c_6) + c_2s_5s_6 = 0$$

$$c_1(c_2c_4s_5 + s_2c_5) - s_1s_4s_5 = 0$$

$$s_1(c_2c_4s_5 + s_2c_5) + c_1s_4s_5 = 1$$

$$-s_2c_4s_5 + c_2c_5 = 0$$

$$c_1s_2d_3 - s_1d_2 + d_6(c_1c_2c_4s_5 + c_1c_5s_2 - s_1s_4s_5) = -0.154$$

$$s_1s_2d_3 + c_1d_2 + d_6(c_1s_4s_5 + c_2c_4s_1s_5 + c_5s_1s_2) = 0.763$$

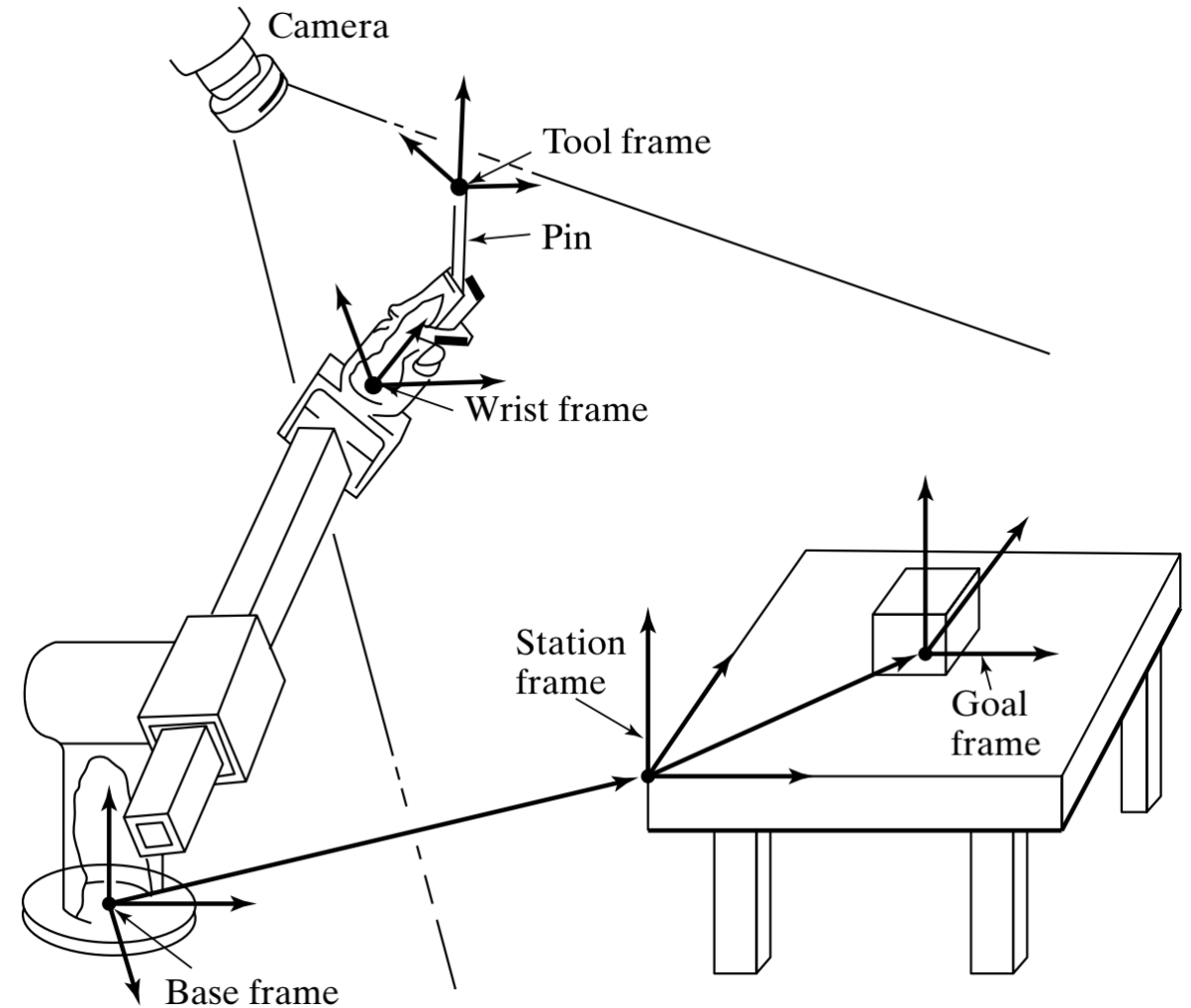
$$c_2d_3 + d_6(c_2c_5 - c_4s_2s_5) = 0$$

$$\theta_1 = \pi/2, \quad \theta_2 = \pi/2, \quad d_3 = 0.5, \quad \theta_4 = \pi/2, \quad \theta_5 = 0, \quad \theta_6 = \pi/2.$$

The Idea of Tool Frame

Workspace also depends on the tool-frame transformation, because it is usually the tool-tip that is discussed when we speak of reachable points in space. Generally, the tool transformation is performed independently of the manipulator kinematics and inverse kinematics, so we are often led to consider the workspace of the wrist frame, $\{W\}$.

If the desired position and orientation of the wrist frame is in the workspace, then at least one solution exists.



Kinematic Decoupling

The general problem of inverse kinematics is quite difficult, it turns out that for manipulators having six joints with three consecutive three joint axes intersecting at a point (such as the Stanford Manipulator above), it is possible to decouple the inverse kinematics problem into two simpler problems, known respectively as inverse position kinematics, and inverse orientation kinematics.

Let us suppose that there are exactly six degrees of freedom, and that the last three joint axes intersect at a point O_c . We express the transformation equations as two sets of equations representing the rotational and positional equations

$$\begin{aligned} R_6^0(q_1, \dots, q_6) &= R \\ o_6^0(q_1, \dots, q_6) &= o \end{aligned}$$

Where o and R are the desired position and orientation of the tool frame, expressed with respect to the world coordinate system. Thus, we are given O and R , and the inverse kinematics problem is to solve for $q_1; \dots; q_6$.

The assumption of a spherical wrist means that the axes $Z3$, $Z4$, and $Z5$ intersect at O_c and hence the origins $O4$ and $O5$ assigned by the DH convention will always be at the wrist center O_c . The important point of this assumption for the inverse kinematics is that motion of the final three joints about these axes will not change the position of O_c , and thus the position of the wrist center is a function of only the first three joint variables.

The origin of the tool frame (whose desired coordinates are given by o) is simply obtained by a translation of distance d_6 along z_5 from O_c . In our case, z_5 and z_6 are the same axis, and the third column of R expresses the direction of z_6 with respect to the base frame. Therefore, we have

$$o = o_c^0 + d_6 R \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$o_c^0 = o - d_6 R \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The orientation of the frame $O_6X_6Y_6Z_6$ with respect to the base be given by R . If the components of the end effector's position O are denoted $O_x; O_y; O_z$ and the components of the wrist center O_c are denoted $X_c; Y_c; Z_c$ then,

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} o_x - d_6 r_{13} \\ o_y - d_6 r_{23} \\ o_z - d_6 r_{33} \end{bmatrix}$$

Now, we may find the values of the first three joint variables. This determines the orientation transformation R_{03} which depends only on these first three joint variables. We can now determine the orientation of the end effector relative to the frame $O_3X_3Y_3Z_3$ from the expression

$$R = R_3^0 R_6^3$$

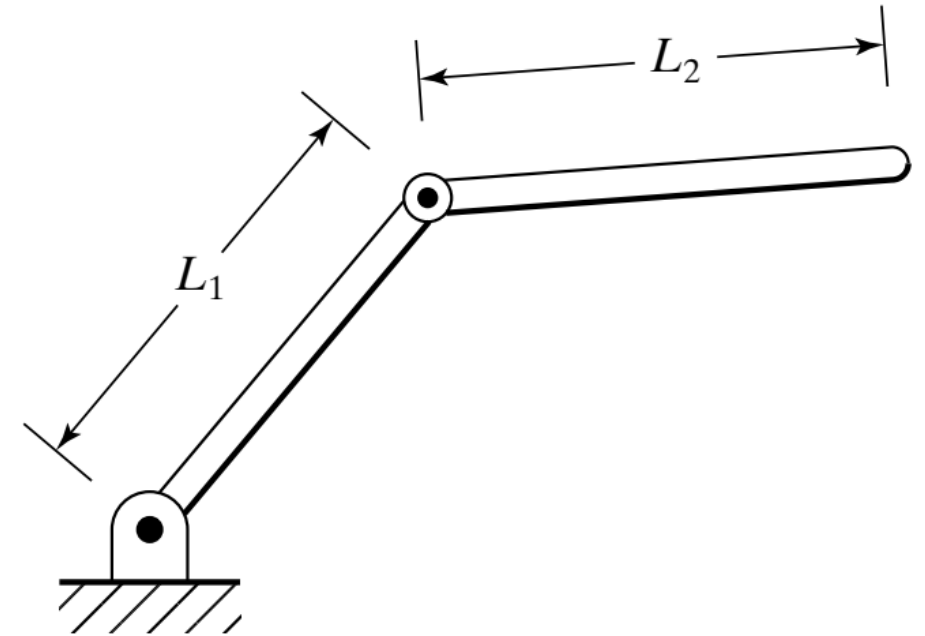
$$R_6^3 = (R_3^0)^{-1} R = (R_3^0)^T R = A_4 A_5 A_6$$

Finally , solve the three equations for the three joint variables.

H.W

Consider the workspace of the two-link manipulator. If $L_1 = L_2$, then the reachable workspace consists of a disc of radius $2L_1$. If $L_1 \neq L_2$, then the reachable workspace becomes a ring of outer radius $L_1 + L_2$ and inner radius $|L_1 - L_2|$. Inside the reachable workspace there are two possible orientations of the end-effector. On the boundaries of the workspace there is only one possible orientation.

These considerations of workspace for the two-link manipulator have assumed that all the joints can rotate 360 degrees. This is rarely true for actual mechanisms. When joint limits are a subset of the full 360 degrees, then the workspace is obviously correspondingly reduced, either in extent, or in the number of possible orientations attainable.



IK Solution Strategies

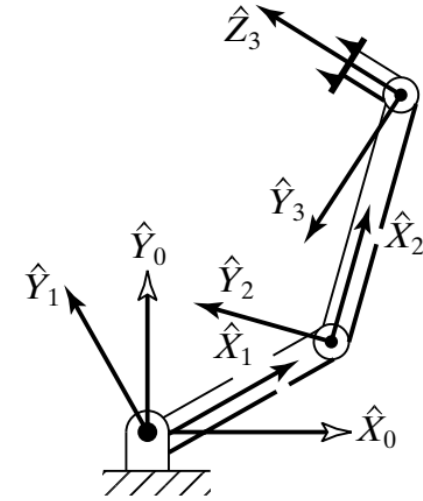
1. Algebraic Method
2. Geometric Method
3. Numerical Method
4. Teach and playback

1. Algebraic Method

Example

Consider the three-link planar manipulator, $\phi = \theta_1 + \theta_2 + \theta_3$

$${}^B_W T = {}^0_3 T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



i	$\alpha_i - 1$	$\alpha_i - 1$	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3

Transformation Matrix :

$${}^B_W T = \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^B_W T = {}^0_3 T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We arrive at a set of four nonlinear equations that must be solved for θ_1 , θ_2 and θ_3 .

$$c_\phi = c_{123}, \quad \cos(\phi) = \cos(\theta_1 + \theta_2 + \theta_3)$$

$$s_\phi = s_{123},$$

$$x = l_1 c_1 + l_2 c_{12},$$

$$y = l_1 s_1 + l_2 s_{12}.$$

$$x = f(\theta_1, \theta_2)$$

$$y = f(\theta_1, \theta_2)$$

Square the last two equations and add them,

$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1l_2c_2$$

$$c_{12} = c_1c_2 - s_1s_2$$

$$s_{12} = c_1s_2 + s_1c_2$$

Re-arranging the equation,

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2},$$

$$\theta_2 = \cos^{-1}(c_2)$$

$$\cos^{-1} \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2},$$

In order for a solution to exist, the right-hand side of the equation (C2) must have a value between -1 and 1. In the solution algorithm, this constraint would be checked at this time to find out whether a solution exists. Physically, if this constraint is not satisfied, then the goal point is too far away for the manipulator to reach.

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}.$$

$$s_2 = \pm \sqrt{1 - c_2^2}$$

$$\theta_2 = \text{Atan2}(s_2, c_2)$$

Re-write the x and y equations as,

$$x = k_1 c_1 - k_2 s_1$$

$$y = k_1 s_1 + k_2 c_1$$

Where:

$$k_1 = l_1 + l_2 c_2,$$

$$k_2 = l_2 s_2.$$

$$c_\phi = c_{123},$$

$$s_\phi = s_{123},$$

$$x = l_1 c_1 + l_2 c_{12},$$

$$y = l_1 s_1 + l_2 s_{12}.$$

Assume:

$$r = +\sqrt{k_1^2 + k_2^2}$$

$$\gamma = \text{Atan2}(k_2, k_1),$$

Then:

$$k_1 = r \cos \gamma,$$

$$k_2 = r \sin \gamma.$$

Hence, x , y may be written in the form:

$$\frac{x}{r} = \cos \gamma \cos \theta_1 - \sin \gamma \sin \theta_1.$$

$$\frac{y}{r} = \cos \gamma \sin \theta_1 + \sin \gamma \cos \theta_1.$$

$$x = k_1 c_1 - k_2 s_1$$

$$y = k_1 s_1 + k_2 c_1$$

Use trigonometry,

$$\cos(\gamma + \theta_1) = \frac{x}{r},$$

$$\sin(\gamma + \theta_1) = \frac{y}{r}.$$

$$\gamma + \theta_1 = \text{Atan2}\left(\frac{y}{r}, \frac{x}{r}\right) = \text{Atan2}(y, x)$$

$$\theta_1 = \text{Atan2}(y, x) - \text{Atan2}(k_2, k_1).$$

Finally,

$$\theta_1 + \theta_2 + \theta_3 = \text{Atan2}(s_\phi, c_\phi) = \phi.$$

$$c_{12} = c_1 c_2 - s_1 s_2$$

$$s_{12} = c_1 s_2 + s_1 c_2$$

$$\frac{x}{r} = \cos \gamma \cos \theta_1 - \sin \gamma \sin \theta_1$$

$$\frac{y}{r} = \cos \gamma \sin \theta_1 + \sin \gamma \cos \theta_1$$

In summary,

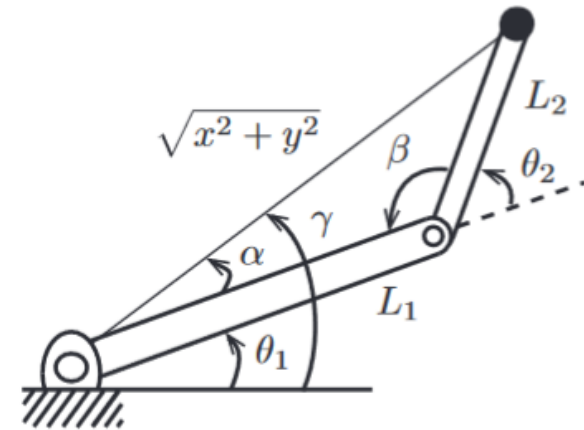
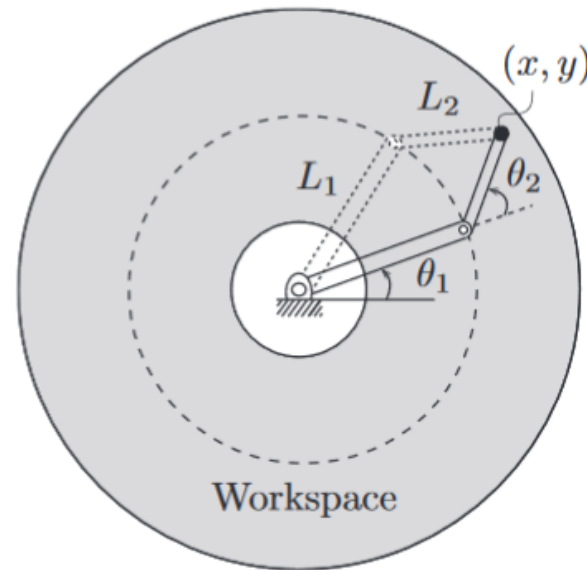
An algebraic approach to solving kinematic equations is basically one of manipulating the given equations into a form for which a solution is known. It turns out that, for many common geometries, several forms of transcendental equations commonly arise.

2. Geometric Solution:

In a geometric approach to finding a manipulator's solution, we try to decompose the spatial geometry of the arm into several plane-geometry problems.

Example: Two link planar open chain

Find (θ_1, θ_2) for a given (x_1, x_2)



2. Geometric Solution, cont.

$$L_1^2 + L_2^2 - 2L_1L_2 \cos \beta = x^2 + y^2$$

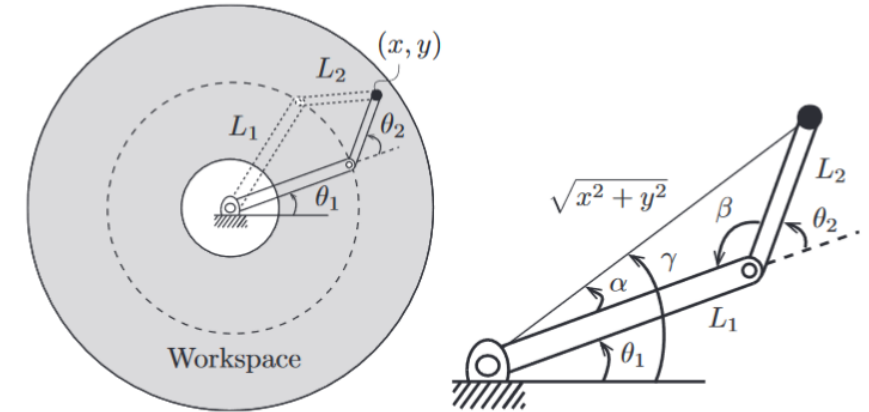
$$\beta = \cos^{-1} \left(\frac{L_1^2 + L_2^2 - x^2 - y^2}{2L_1L_2} \right)$$

$$\alpha = \cos^{-1} \left(\frac{x^2 + y^2 + L_1^2 - L_2^2}{2L_1\sqrt{x^2 + y^2}} \right)$$

$$\theta_1 = \gamma - \alpha, \quad \theta_2 = \pi - \beta$$

$$\theta_1 = \gamma + \alpha, \quad \theta_2 = \beta - \pi$$

where, $\gamma = \text{atan2}(x, y)$



Law of cosine

$$c^2 = a^2 + b^2 - 2ab \cos(C)$$

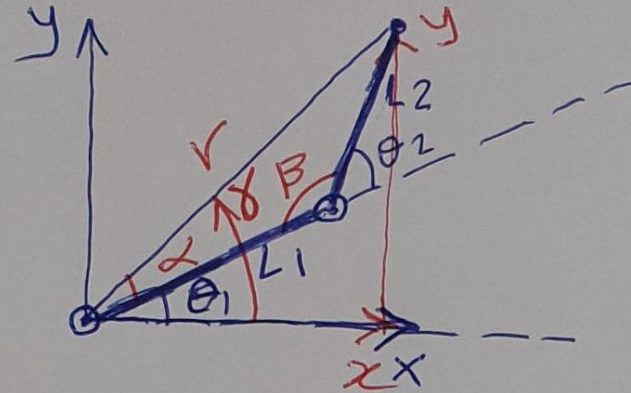
Where a, b, and c are the lengths of the three sides of a triangle and C is the interior angle of the triangle opposite the side of length c.

Example:

Law of cosine

$$c^2 = a^2 + b^2 - 2ab \cos(C)$$

Where a, b, and c are the lengths of the three sides of a triangle and C is the interior angle of the triangle opposite the side of length c.



and, $r^2 = x^2 + y^2$

$$r^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos \beta$$

(from cosine rule)

or,

$$\cos \beta = \frac{L_1^2 + L_2^2 - r^2}{2L_1L_2}$$

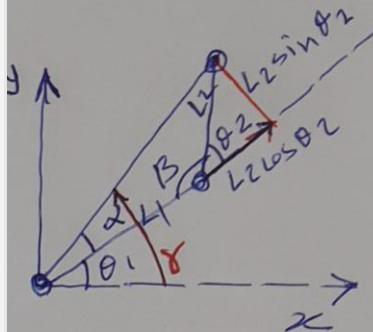
$$\cos \beta = \frac{L_1^2 + L_2^2 - x^2 - y^2}{2L_1L_2}$$

$$\theta_2 = \pi - \beta$$

$$\therefore \cos \theta_2 = -\cos \beta$$

Example, cont.

$\theta_2 = \pi - \beta$
 $\therefore \cos \theta_2 = -\cos \beta$
 $\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2}$
 $\theta_2 = \cos^{-1} \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2}$

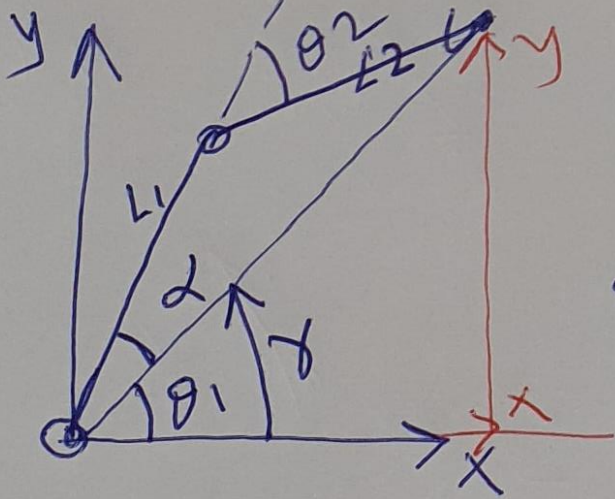


$L_2^2 = L_2^2 \sin^2 \theta_2 + L_2^2 \cos^2 \theta_2$
 $\sin \theta_2 = \sqrt{1 - \cos^2 \theta_2}$
 $\alpha = \tan^{-1} \frac{L_2 \sin \theta_2}{L_1 + L_2 \cos \theta_2}$
 $\gamma = \tan^{-1} \frac{y}{x}$
 and,
 $\theta_1 = \gamma - \alpha$

$$\theta_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{L_2 \sin \theta_2}{L_1 + L_2 \cos \theta_2}$$

Example, cont.

Also, $\cos \theta = \cos(-\theta)$

$$\therefore \theta_2 = -\cos^{-1} \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2}$$


$$\gamma = \tan^{-1} \frac{y}{x}$$

$$\theta_1 = \gamma + \alpha$$

$$\theta_1 = \tan^{-1} \frac{y}{x} + \tan^{-1} \frac{L_2 \sin \theta_2}{L_1 + L_2 \cos \theta_2}$$

Taught and Computed Points, Repeatability and Accuracy



Many industrial robots today move to goal points that have been taught. A **taught point** is one that the manipulator is **moved to physically**, and then the joint position sensors are read and the joint angles stored. When the robot is commanded to return to that point in space, each joint is moved to the stored value. In simple "teach and playback" manipulators such as these, the inverse kinematic problem never arises, because goal points are never specified in Cartesian coordinates. When a manufacturer specifies how precisely a manipulator can return to a taught point, he is specifying the **repeatability** of the manipulator.

- Any time a goal position and orientation are specified in Cartesian terms, the inverse kinematics of the device must be computed in order to solve for the required joint angles. Systems that allow goals to be described in Cartesian terms are capable of moving the manipulator to points that were never taught—points in its workspace to which it has perhaps never gone before. We will call such points **computed points**. Such a capability is necessary for many manipulation tasks. For example, if a computer vision system is used to locate a part that the robot must grasp, the robot must be able to move to the Cartesian coordinates supplied by the vision sensor. The precision with which a computed point can be attained is called the **accuracy** of the manipulator.



Analytical solutions

- Write down lots of equations
- Needs some human insights
- Can be automated
- Explicitly shows multipole configuration
- Solutions are compact and fast to execute
- Increasingly difficult as the number of joints increase

3. Numerical Method

Example: Newton-Raphson method

Start

Input: final pose (x,y,z)

*Build the Transformation matrix between the initial
and desired (final) pose*

Initialize the joint angles, Guess

*Set the mask matrix (rotation and orientation
references)*

Optimization algorithm

Output: joint angles

End

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

4. Teach and Playback



<https://www.youtube.com/watch?v=tub6oUeMhqk>

Questions?